CDAC CCEE: Data Visualization - Quick Revision Cheat Sheet

1. The Data Analytics Lifecycle (Memorize Order)

Examiners ask for the sequence or "what comes next".

1. **Discovery:** Assess resources, understand business problem, acquire domain knowledge.
2. **Data Preparation:** ETL, cleaning, transformation (Structured vs. Unstructured).
3. **Model Planning:** Select methods/algorithms, explore variables (variables selection).
4. **Model Building:** Execute the model (coding/implementation).
5. **Quality Assurance (QA):** Test validity and accuracy. *Critical step before deployment.*
6. **Documentation:** Record process/logic for reproducibility.
7. **Management Approval:** Stakeholder sign-off.
8. **Installation/Deployment:** Operationalization/Go-live.

2. Chart Taxonomy (Which Chart to Use?)

Key: Match the "User Goal" to the "Chart Type".

User Goal	Recommended Charts	Keywords / Notes
Comparison (Categorical)	Bar Chart, Side-by-Side Bar	Comparing values across categories.
Comparison (Performance)	Bullet Graph	Actual vs. Target (Replaces Gauges).
Distribution	Histogram, Box Plot	**Box Plot:** Shows outliers & quartiles. **Histogram:** Frequency bins.
Part-to-Whole	Pie Chart, Stacked Bar, Tree Map	**Tree Map:** Nested rectangles for hierarchy.
Trends over Time	Line Chart, Area Chart	**Area:** Shows volume/magnitude over time.
Relationships	Scatter Plot, Heat Map, Bubble Chart	**Scatter:** Correlation (X vs Y).

		Heat Map: Density/Intensity (Color).
Project Schedule	Gantt Chart	Start Date + Duration.
Geospatial	Symbol Map, Filled Map	Latitude/Longitude data.

3. Visual Encodings & Theory

- **Visual Encodings:** The ways data is mapped to visual structures.
  - **Most Accurate (Quantitative):** Position (e.g., Bar/Scatter).
  - **Least Accurate (Quantitative):** Color hue, Shape.
- **Visual Analytics:** Integrating *automated analysis* with *interactive visualizations* for human reasoning.
- **Structured Data:** RDBMS, Excel, CSV (Rows/Columns).
- **Unstructured Data:** Video, Audio, Images, Free text.

4. Tableau Specifics (Technical Terms)

- **Dimensions vs. Measures:**
  - **Dimensions:** Qualitative (Dates, Names, Regions). Slices the data.
  - **Measures:** Quantitative (Sales, Profit). Aggregates the data.
- **Blue vs. Green Pills:**
  - **Blue: Discrete** (Creates Headers).
  - **Green: Continuous** (Creates Axes).
- **LOD (Level of Detail):**
  - Allows calculations at a different granularity than the view (FIXED, INCLUDE, EXCLUDE).
- **Dashboards:**
  - **Action Filters:** Clicking a chart filters other charts.
  - **Tiled vs. Floating:** Layout options for arranging objects.

5. MS Excel for Analytics

- **Function vs. Formula:**
  - **Formula:** User-defined equation (e.g., `=A1+B1`).
  - **Function:** Built-in keyword (e.g., `=SUM(A1:B1)`).

- **Pivot Tables:** Best for summarizing/aggregating large datasets and Cross-tabulation.
- **Data Analysis Toolpak:**
  - **Descriptive Statistics:** Summary (Mean, Median, Mode, SD).
  - **Correlation:** Strength of linear relationship (-1 to +1).
  - **Regression:** Predicting trends (fitting a line).
  - **VLOOKUP:** Vertical Lookup (Needs lookup value in the *first* column).